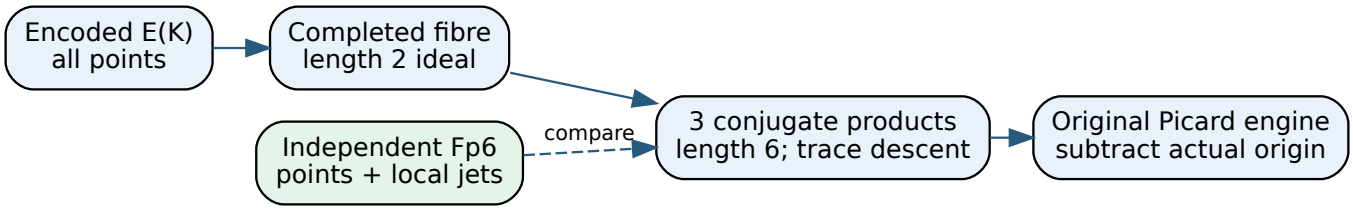
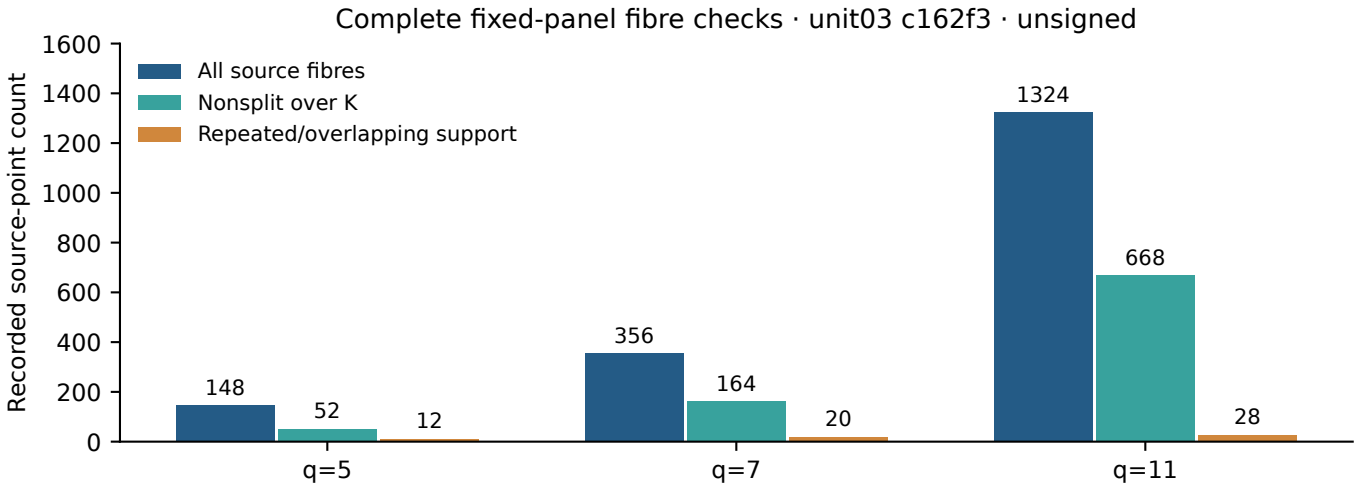
Completed prime-field genus-three descent

Native component result · 2026-11-13 · local acceptance / unsigned





The completed encoded-point map passes all four native tests on the original fixed q=5,7,11 panel: every source point, complete divisor scheme, generating edge and odd-subgroup scalar image. Unit03 c162f3 (whole custody2c86a8) has original native0, final guard0, all811 hashes unchanged, zero ignored or filtered tests. The source domains contain148,356,1324 points. The complete generating-edge counts are592,1424,5296; odd scalar counts37,89,331; observed even kernel2 in each case. These are unsigned local observations. [pcc:unverified]

Runtime construction: a completed length-two fibre ideal (L,Q), the product of its three conjugate Cartier ideals, coefficient traces over1,g,g², then the original canonical Picard completion/reduction and subtraction of the actual origin fibre. The origin has multiplicities3,1,1,1. Runtime input is an encoded point with exact field/model tags; no support roots, scalar labels or fitted origin constant. The separate FpR6 point-and-local-jet oracle checks cubic spaces and reduced classes for every source point. [pcc:unverified]

q	K-rational branches	Geometric branches	Origin multiplicities
5	0	4	1,1,1,3
7	4	4	1,1,1,3
11	4	4	1,1,1,3

Preserved failures and corrected coverage

The first native attempt7f48f6 rejects all four tests before point checking: the historical quartic is not merely a scalar multiple in the assumed coordinates. The original builder uses primitive gamma=g+s, while the Diem core uses polynomial g. Frozen corrective6a3cfe precedes the exact triangular Moore transport. Unit03 records s=4,1,0 and polynomial scale1,2,1 for q=5,7,11. The entire quartic polynomial identity is checked before using the exact historical coefficients; the independent extension oracle constructs gamma rows directly. [pcc:unverified]

Unit02 b8e858 passes coordinates, complete group and strict controls, and all148 q=5 fibre comparisons, then fails the assumption ramified>0. The frozen all-nine panel6b2a53, run69e6a0, finds primary K-rational ramification counts0,4,4. At q=5 the two u1 roots give u0*u2 norms3 and2 in F5, both nonsquares, so the four geometric branch images lie outside K. All primary fixtures and original failures remain. [pcc:unverified]

Coverage addendum d664a3 is frozen before the next edit/run. Unit03 checks the exact rational counts0,4,4 and adds all four geometric branch images per case over Fp6. Each actual plane-curve local jet has X of order one and cover t-t0 of order two, with Q=X² of order two and L vanishing modulo the squared local parameter. Thus the fibre keeps its doubled-point structure. All12 branch controls pass. This does not extend the public runtime beyond E(K). [pcc:unverified]

G3_PROOF_MAP.md/e6590f identifies the actual V4 quotient argument: diagonal norm/pullback[2], off-diagonal zero by character separation and torsion-free Hom, descent from Res(K/Fq)E0, degree8, and odd-subgroup injection by Bezout. Original Lean4.35 accepts nine Descent statements plus three generated helpers (d9f9f6) and two ModelTransport statements (5cbc69), zero trusted declarations. All eleven exact original PCC bindings locally pass. Actual curve/Picard/Cartier/descent/reducer identifications remain written/native bridges, rather than formal definitions. [pcc:unverified]

Three different mathematical scopes

The runtime fibre has degree2; its rational trace divisor has degree6; the descended product isogeny has degree8. Finite native enumeration checks fixed encoded groups. The Lean files prove the exact algebra and conditional group implications listed in the proof map. The written geometry identifies those hypotheses with the actual covers. [pcc:unverified]

Execution, publication and reproducibility

Strict static history:01=101 on original dependency warnings (604a22);02=101 on one unused new test binding (65bae1);03=0 (8c3b6a); corrected-source04=0 (0a50c9). Compatibility allowances match original imports; original arithmetic bodies and new-source warning criteria remain. Lean01 fails a generic symbolic-smul tactic (224f01); Lean02 proves the identical statements with explicit module laws. Metadata parser/serializer failures72ef74,b7fecf are separate, retained observations. [pcc:unverified]

The bounded native CLI188d44 (whole custody6849fe) passes three literal requests/nine points (O, finite x=0 and negative infinity), eight native exit2 controls, unchanged output on overwrite, and all820 final guards. CLI acceptance is bounded; the all1828-point domain belongs to unit03. Controls reject foreign field/model, noncanonical/off-curve/domain/fixture/schema/size mutations in the native suite. [pcc:unverified]

Actual publication0775c2: trusted-base run43370167545/job131703822791 replays FOUR whole remote objects before normal manifest botf95c2ab, parent eab262. These are static01/02/03 and failed unit01. All object sizes/hashes match intact originals; exact local fast-forward, artifact check and wiring check exit0. The whole427120-byte UTF8 log and whole26-file publication tree are retained; publication-tree XZ74964 bytes will itself use the genuine artifact route. Latest unit02/unit03/static04/CLI whole raw handoffs are separately pending remote publication. [pcc:unverified]

Reproducibility: Rust1.99.1; original Lean4.35; local Linux x86_64 correctness commands; fresh build caches outside raw boundaries. Original commands, source snapshots, stdout/stderr, exits and final guards are whole standard tar/XZ archives with member/raw/archive hashes and exact roundtrip. No build duration is a performance result. New focused CI contract6b95c7 preserves the existing root four-job and consumer six-job workflows; local integrated replay and root library regression are separate pending gates at report preparation. [pcc:unverified]

Primary sources: G3_DESCENT_PROTOCOL.md and DERIVATION, evidence/genus3-primary-01/sources.json; original ghs_cubic_genus3_build_20261012/genus3_build.rs receipt a10f0d2d; original diem_quartic.rs and plane_quartic_pic.rs; original isogeny_algos FpR/curve/poly/series APIs; cited Diem and Joux–Vitse primary papers. KB unavailability is recorded, not evidence of novelty. Existing C0/T20 graphs and cost tables are unchanged. Next: finish focused integration/CI and independent consumer replay, then the arbitrary-Jacobian norm and destination index-calculus pipeline. A charged computational comparison still requires actual admission. [pcc:unverified]